

Qatar-2b

R-1966

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_130409 · created 2026-08-06T23:20:57+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456391.77177 +/- 0.00027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.093 +/- 0.067
Transit depth (Rp/Rs)^2	0.86 +/- 1.24 [%]
Orbital Inclination (inc)	86.91 +/- 1.74
Airmass coefficient 1 (a1)	0.66 +/- 0.35
Airmass coefficient 2 (a2)	0.33 +/- 0.48
Scatter in the residuals of the lightcurve fit is	54.92 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.0677 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.093 ± 0.067 vs 0.16327 (deviation 1.05σ)
Duration fit vs published	0.0677 d vs 0.0767 d (deviation 1.34σ)
Mid-time O–C	-0.4 min
Depth SNR	1.4
Residual scatter	54.92 %

Light curve

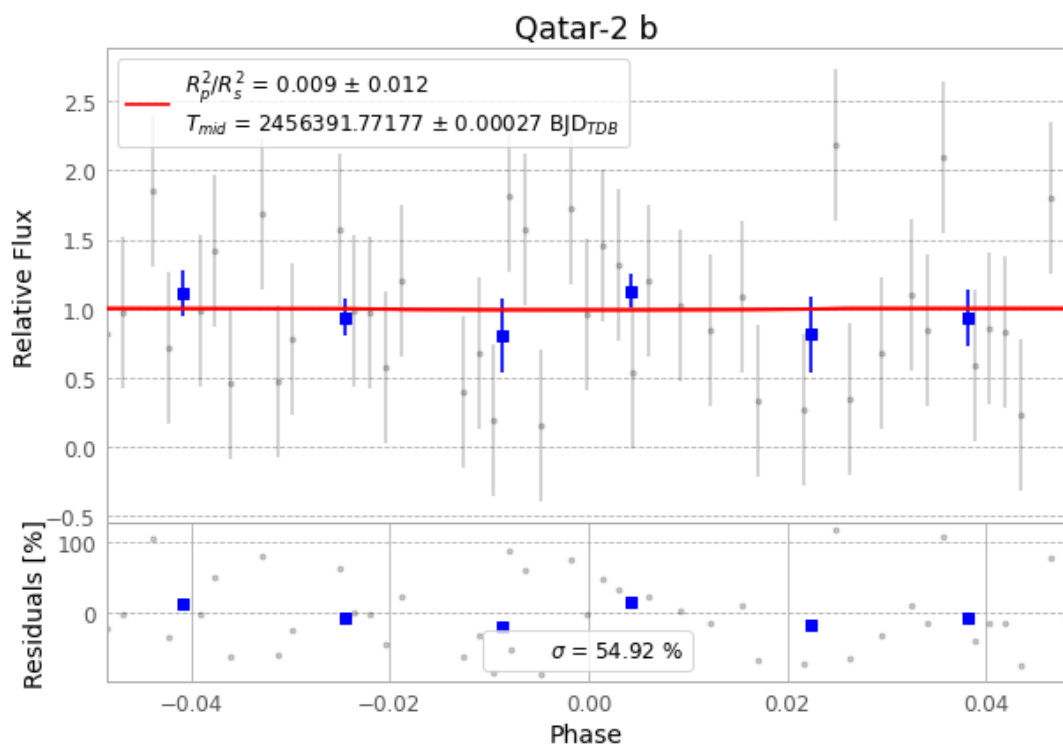


Figure 1 — FinalLightCurve_Qatar-2 b_2013-04-08.png (SHA-256 6000a1fb39ccdc2c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [589, 173], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 ad3ca0c00a7f1782...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

62 FITS frames (41 MB), Qatar-2130409044812.FITS → Qatar-2130409075712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-130409.log (d0d52fadcf8...), FinalLightCurve_Qatar-2 b_2013-04-08.png (6000a1fb39cc...), FinalLightCurve_Qatar-2 b_2013-04-08.csv (d626cbc9a5df...)

Compute resources

Wall time 113.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 23:20Z.